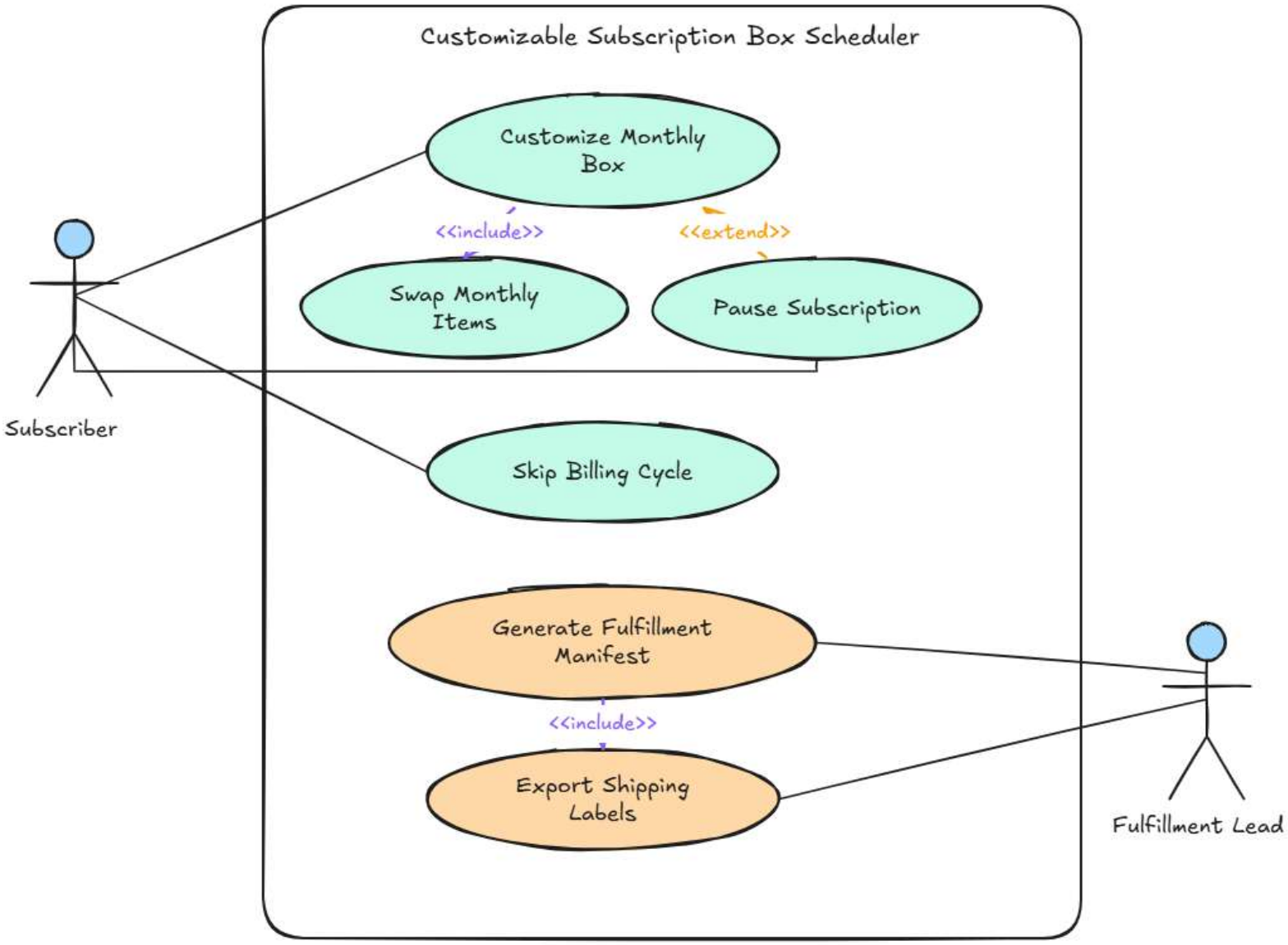
Requirements Table

Customizable Subscription Box Scheduler

Req ID	Type	Priority	Description	Acceptance Criteria	Rationale
FR-001	Functional	High	The system shall allow subscribers to customize monthly product selections using preference tags.	The subscriber can choose or update items for the monthly box, and the system saves the selection successfully.	This is the core purpose of the subscription box portal.
FR-002	Functional	High	The system shall allow subscribers to swap monthly box items up to 48 hours before the monthly billing renewal date.	Swaps are accepted before the cutoff and rejected after the cutoff.	This matches the customization rule given in the problem statement.
FR-003	Functional	High	The system shall allow subscribers to pause deliveries before the monthly billing renewal date.	The subscription is marked as paused for the next cycle when the request is made before the deadline.	This is directly supported by the problem statement.
FR-004	Functional	Medium	The system shall allow subscribers to skip one billing cycle or monthly delivery before the renewal cutoff.	The selected cycle is skipped and no delivery is prepared for that cycle.	The handout explicitly mentions skipping billing cycles.
FR-005	Functional	High	The system shall allow the Fulfillment Lead to generate the monthly fulfillment manifest and export shipping labels.	The Fulfillment Lead can generate the manifest and export labels for the monthly batch successfully.	The fulfillment role and shipping-label export are part of the given scenario.
NFR-001	Non-Functional	High	The fulfillment manifest generator shall export shipping labels for 10,000 monthly boxes in under 1 minute.	A benchmark test with 10,000 boxes completes label export within 60 seconds.	This is the sample performance requirement given in the handout.
NFR-002	Non-Functional	High	The system shall protect subscriber and subscription data from unauthorized access.	Unauthorized users cannot view or modify subscriber data, and access is restricted by login or role checks.	This matches the security-oriented NFR guidance in the lab.

Note: The wording above keeps to the lab brief and avoids inventing automatic payment details that are not stated in the handout.


Customizable Subscription Box Scheduler
Customize Monthly Box
<<include>>
<<extend>>
Swap Monthly Items
Pause Subscription
Skip Billing Cycle
Generate Fulfillment Manifest
<<include>>
Export Shipping Labels
Subscriber
Fulfillment Lead

Use-Case Flow: Customize Monthly Box

Use Case

Customize Monthly Box

Goal

Allow a subscriber to update the monthly box selection before the renewal cutoff, so the finalized order can later be used for fulfillment. The lab also requires one use-case flow with a main success scenario and at least one alternate flow.

Preconditions

- The subscriber is registered and has access to the subscription box portal.
- The monthly box for the upcoming cycle is open for editing.
- The current time is still before the cutoff window, which the handout describes as up to 48 hours before the monthly billing renewal date.

Postconditions

- The subscriber's monthly box selection is saved successfully.
- The updated selection is available for final processing before fulfillment.
- If the cutoff has passed, the selection remains locked and no further change is accepted.

Main Success Scenario

1. The subscriber opens the subscription box portal.
2. The system displays the current monthly box and available preference-based product options.
3. The subscriber selects items to add, swap, or update.
4. The subscriber confirms the new monthly selection.
5. The system checks whether the request is within the allowed editing window.
6. The system saves the updated selection.
7. The system shows a confirmation message.
8. The finalized selection remains available for downstream fulfillment processing.

Alternate Flow

A1: Subscriber tries to edit after the cutoff 1. The subscriber opens the monthly box after the editing deadline. 2. The system checks the renewal cutoff date. 3. The system rejects the change request. 4. The system displays a message that the monthly selection is locked for the current cycle.